# LEONARDO FERREYRA CANAVAL

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## Professional Summary

Software Engineer with 4+ years of experience designing scalable backend systems, AI-driven architecture, and cloud-native solutions. Specializes in Java/Spring Boot, Python, Node.js/NestJS, RESTful APIs, and distributed systems, with deep expertise in AI agent frameworks (CrewAI, LangGraph), LLM observability (Phoenix/Arize), and MCP-based automation. At UPC, led the rebuild of the university's ABET accreditation platform as main developer on a team of 7, owning the NestJS/PostgreSQL architecture, the data model, the authenticated scraping subsystem, and the GitHub Actions CI/CD now running it in production. At Kinara Systems, designed and delivered multi-agent AI pipelines for real-time network analytics, integrating self-hosted and cloud LLMs, building LLM proxy infrastructure, and driving traceability with Phoenix. At Enviame, improved system reliability through Redis-based batching and automated developer workflows via MCP integrations with JIRA, Confluence, and Bitbucket. Proficient in Node.js, Python, Docker, Kubernetes, Redis, AWS, and GCP. Passionate about building intelligent, observable, and production-ready AI systems.

## Skills

**AI & Agents:** CrewAI, LangGraph, MCP (Model Context Protocol), RAG Pipelines, LLM Orchestration, Phoenix/Arize (LLM Tracing & Observability)

**Backend:** Java / Spring Boot, Python (Flask, Django), Node.js (NestJS, Express), .NET / C#, TypeORM, RESTful APIs, OAuth2 / JWT Auth, Microsoft Entra ID (MSAL), Redis, MQTT

**Frontend:** React / Redux, Next.js (App Router), Angular, Vue.js, Flutter, TypeScript, Tailwind CSS, TanStack Query, Zod

**Cloud & DevOps:** GCP (Cloud Run, Cloud Functions, Firestore), AWS (RDS, S3), Docker & Docker Compose, Kubernetes, CI/CD, GitHub Actions, GHCR, Bitbucket Pipelines

**Databases:** PostgreSQL, MySQL, SQL Server, InfluxDB, Firestore, Redis

**Automation:** Playwright, Puppeteer (authenticated scraping, headless and streamed headful browsers)

**Methodologies:** Domain-Driven Design, MVC, OOP, Agile / Scrum, Clean Architecture

**Languages:** Spanish (Native), English (Professional Working, B2 TOEFL)

## Work Experience

**Lead Software Engineer** | **UPC** | Lima, Peru (Remote) *03/2025 – 08/2026*

- Principal engineer of ABET 3.0, the university's accreditation platform now in production, on a team of 7. Led the rebuild of a legacy .NET / SQL Server system into a NestJS 11 + TypeORM + PostgreSQL backend (77 entities, 81 migrations, 300 test suites) and a Next.js 16 / React 19 / TanStack Query / Zod frontend, defining the core architecture and data model.
- Cut the RC grades export from 26 minutes to 85 seconds (18x) by profiling the production query with EXPLAIN ANALYZE, and made PDF generation 4x faster (430 ms to ~100 ms per file) by moving the renderer to a persistent Chromium with concurrent pages.
- Built the machine-to-machine integration API consumed by the university's portfolio audit system, covering IFC PDF access, grade bands, evidence status, report manifests, bulk ZIP exports, and issuance, rotation, and revocation of integration keys.
- Built a real PostgreSQL integration test harness on a production schema dump, covering the 27 plpgsql upload and rollback functions, after finding defects that had shipped past a green suite of 2,000+ mocked tests.
- Designed the team's spec-driven change workflow (proposal, design with an ADR gate, tasks, runbook) and authored its 15 ADRs and repo-wide policies, plus Claude Code skills that promote releases through protected branches.
- Created abet-plugins, the team's AI assisted development standard, built from one shared source of skills, agents, and policy rules that generates plugins for Claude Code, Codex, and opencode. The same checks are wired twice, as PreToolUse hooks that a --no-verify flag cannot bypass and as husky git hooks for the other providers, with 425 test cases over the rule set. Adopted by all 7 developers on the team.
- Owned the IFC module (end of cycle course report) and the ARD module (class representative meeting minutes) end to end across backend and frontend, covering the state machine (Saved, Submitted, Approved, Observed), authorization derived from the organizational chart rather than fixed roles, findings and improvement actions, email notifications, and PDF / Excel reporting. Authored the IFC technical documentation and end user manual.

- Owned the data loading domain, 14 bulk upload pipelines (sections, enrolled students, RC and RV grades, outcomes, study plans, staff, rubrics, projects, class representatives, organizational charts) with Excel parsing, row level validation, upload logs, and rollback.
- Built the Banner and Planner scraping subsystem with Playwright and Puppeteer, including a headful Chromium streamed to the operator over noVNC so staff complete the MFA login live while the scraper reuses the session, later extracted into a shared browser service.
- Designed the CI/CD from scratch with GitHub Actions (CI gate that deploys cannot bypass, cached Docker images on GHCR, SSH deploy via Docker Compose, on demand migrations, image rollback), and consolidated the backend and frontend repos into a pnpm monorepo.
- Hardened authentication on top of Microsoft Entra ID SSO, adding CSRF protection and school validation to the login and the service that scopes every request to the user's school, and encrypted external-system credentials at rest.
- Previously, on ABET 2.0, maintained the legacy .NET / C# clean architecture backend and React frontend, owning the load maintenance module (CRUD over 10 loaded entity types plus load rollback and retry, ~12k LOC) and acting as production support for the accreditation system.

**AI Software Engineer** | **Kinara Systems** | Canada (Remote) *11/2025 – 04/2026*

- Designed and delivered a multi-agent AI pipeline for real-time Wi-Fi network analytics, leveraging CrewAI and LangGraph to orchestrate specialized agents for telemetry analysis, log classification, ping-pong detection, and sticky-client diagnosis.
- Architected and implemented an LLM proxy layer supporting dynamic routing between self-hosted models (Qwen, DeepSeek, ZGLM) and cloud providers (OpenAI, Anthropic), enabling seamless model swapping and A/B evaluation across the system.
- Integrated Phoenix (Arize) for end-to-end LLM observability and tracing, capturing MCPClient calls, CrewAI agent sessions, and tool-call execution to identify latency bottlenecks, hallucination patterns, and model inconsistencies across providers.
- Designed and built the Streaming Telemetry Analyzer and Streaming Log Analyzer services, Python backends that consume MQTT-based network data, apply LLM-driven classification, and expose runtime-configurable APIs for dynamic threshold and pattern updates without service restarts.
- Built configuration management APIs for both analyzers with YAML-based persistence, enabling runtime updates to LLM parameters, alert thresholds, and monitoring patterns, validated via full integration test suites.
- Evaluated and benchmarked multiple LLMs (GPT-4o, GPT-OSS-120B, DeepSeek V3, Qwen3 235B, ZGLM 4.6) on real network queries, documenting tool-call behavior, latency profiles, and output quality to drive model selection decisions.
- Implemented a TTS (Text-to-Speech) proxy integration across both analyzers, including environment variable propagation, header handling, license-gated activation, and metrics formatting, and coordinated alignment across backend services and UI.
- Developed scale-testing infrastructure using InfluxDB and Mosquitto (Docker), generating synthetic datasets of up to 100 APs with dummy client telemetry to stress-test agent pipelines and validate detection logic under load.
- Built the sauble-microagents configuration module and refactored the service to consume client state from the telemetry service, decoupling detector responsibilities and improving system maintainability.
- Implemented license enforcement across the AI pipeline, integrating claim-token validation, license-gated service termination, and end-to-end testing against deployed license APIs.
- Collaborated with cross-functional teams via Confluence documentation, Jira sprint planning, and code reviews, delivering production-grade deployments with thorough regression testing and root-cause analysis workflows.

**Software Developer** | **Enviame** | Chile (Remote) *04/2025 – 11/2025*

- Implemented backend features end to end on Google Cloud (Cloud Run, Cloud Functions, Firestore, Cloud Storage), from design to delivery.
- Designed and optimized MySQL data access patterns, resolving critical performance issues caused by inconsistent data types and inefficient queries, resulting in significantly faster response times.
- Improved system reliability by implementing a Redis-based batching system to regulate outbound requests, reducing rate-limit errors (HTTP 429) and stabilizing external API communication.
- Automated the internal developer workflow by integrating MCP with JIRA, Confluence, and Bitbucket, covering API configuration, workflow orchestration, and automatic PR generation, which reduced manual work and improved team productivity.
- Built and maintained backend services using Python (Flask), focusing on clean architecture, maintainability, and performance.

**Software Developer** | **CIVA** | Lima, Peru *04/2024 – 04/2025*

- Developed backend for the FLIT transportation ERP using Java/Spring Boot, implementing JWT authentication and RESTful APIs.

- Optimized response times by implementing Data Transfer Objects (DTOs), achieving a 30% reduction in latency.
- Migrated from basic authentication to JWT with Spring Boot Security, enabling integration with Microsoft Azure SSO.
- Developed frontend components with React/Redux following Domain-Driven Design principles within Agile/Scrum workflows.

**Software Developer** | **SOLYMAN Consultores de TI** | Lima, Peru (Remote) *09/2023 – 02/2024*

- Developed a currency and cryptocurrency exchange platform in Odoo using MVC and OOP with real-time REST API integrations.
- Implemented dynamic JavaScript visualizations for trend analysis and a PostgreSQL-backed reporting module for SBS.

**Software Engineer** | **VALENTOR Producciones S.A.C.** | Lima, Peru (Remote) *04/2022 – 09/2023*

- Built the personnel management system (shift scheduling, payroll, role-based access) as part of a team, with visibility across the full stack: Django backend (models, REST API, admin panel, authentication) serving a Flutter mobile app for staff and an Angular web panel for administrators.
- Developed the attendance control system end-to-end on Vue.js (frontend) and .NET (backend), integrating Azure AD for authentication and a third-party facial recognition library to automate clock-in/clock-out.
- Implemented cloud infrastructure on AWS with CI/CD, Docker, and Kubernetes, improving deployment speed by 60% and significantly reducing downtime.

## Education

**Master in Digital Business** | **MIOTI** | Madrid, Spain (Online) *10/2026 – 2027*

**B.Sc. Software Engineering** | **Universidad Peruana de Ciencias Aplicadas** | Lima, Peru

*GPA: 17.95/20 • Consistently ranked top 5 from 1st–10th semester*

**English Language Studies (B2 TOEFL)** | **Instituto Cultural Peruano Norteamericano** | Lima, Peru

## Certifications

- Generative AI Fundamentals Specialization, IBM (2025)
- AI Agents, Platzi (2025)
- NestJS, DevTalles (2025)
- Next.js, DevTalles (2025)
- Full Stack MERN Development, Udemy (2024)
- Node.js, DevTalles (2024)
- Microsoft Azure, Microsoft (2023)